Final Lecture: Infinite Relational Model

Jesse Mu

2016-11-02

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1 The Infinite Relational Model

What happens when you want to cluster your data, but the number of clusters is unknown? While some approaches involve fitting several models with a varying number of clusters to your data and comparing model fit statistics, the Bayesian approach is to specify a model which is allowed to dynamically grow the number of clusters as the complexity of the data warrants.

The **Infinite Relational Model** is the prototypical example of such a model (Kemp et al. 2006).

For this last project, I coded up a simple version of Charles Kemp's Infinite Relational Model (IRM) in `irm.R` to co-cluster rows and columns of a simple 2-dimensional binary relation.

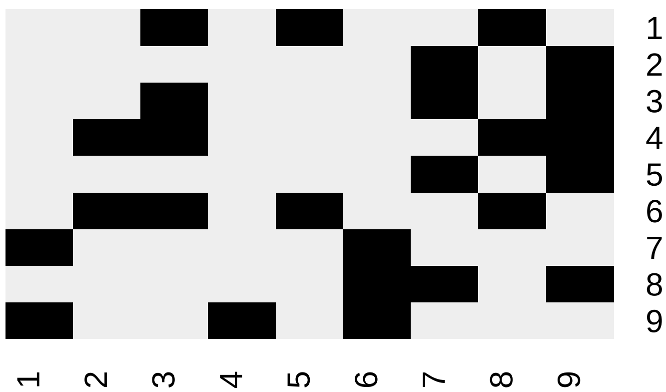
1.1 Demo

```r
source('irm.R')
```

1.1.1 Sanity check

As a sanity check, we use the toy matrix in the original paper:

```r
R = rbind(
  c(0, 0, 1, 0, 1, 0, 0, 1, 0),
  c(0, 0, 0, 0, 0, 0, 1, 0, 1),
  c(0, 0, 1, 0, 0, 0, 1, 0, 1),
  c(0, 1, 1, 0, 0, 0, 0, 1, 1),
  c(0, 0, 0, 0, 0, 0, 1, 0, 1),
  c(0, 1, 1, 0, 1, 0, 0, 1, 0),
  c(1, 0, 0, 0, 0, 1, 0, 0, 0),
  c(0, 0, 0, 0, 0, 1, 1, 0, 1),
  c(1, 0, 0, 1, 0, 1, 0, 0, 0)
)
plot.R(R)
```




```
Z = irm(R, sweeps = 1000)
top.n(Z)
#> 122121323 122121324 122321424 122123424 122323424 122321425 123121424 122324525
#>       884        37        29        18        15         8         5         2
#> 123451637 123456758
#>         1         1
plot.R(R, mode.irm(Z))
```

This successfully finds the clusters of rows and columns that correspond to the original paper.

Kemp, Charles, Joshua B. Tenenbaum, Thomas L. Griffiths, Takeshi Yamada, and Naonori Ueda. 2006. "Learning Systems of Concepts with an Infinite Relational Model." *Proceedings of the National Conference on Artificial Intelligence (AAAI)* 21: 381–88. http://web.mit.edu/cocosci/Papers/Kemp-etal-AAAI06.pdf.